

UWB Localization $2\text{m} \times 2\text{m}$ Test Report

Objective

Evaluate the accuracy and stability of a three-anchor UWB localization system within a $2\text{m} \times 2\text{m}$ test area and identify major sources of localization error.

Experimental Setup

Three anchors were placed at:

- $A1 = (0,0)$
- $A2 = (0,2)$
- $A3 = (2,0)$

The tag was moved along a “+” shaped trajectory while recording range measurements and estimated coordinates. Position estimation was performed using trilateration followed by EKF-based filtering.

Observations

- The estimated trajectory did not accurately follow the expected “+” pattern.
- Initially, the issue appeared to be related to boundary effects; however, errors were present throughout the workspace.
- At the center position $(1, 1)$, the estimated coordinate was approximately $(1.037, 0.888)$.
- For a tag located at $(1, 1)$, the distance to all three anchors should be approximately 1.414 m .
- Measured ranges deviated significantly from the expected values.
- Human movement around the setup did not noticeably affect localization performance.
- Communication latency was not identified as a major issue.
- X-coordinate estimates were relatively stable, whereas Y-coordinate estimates showed larger fluctuations.

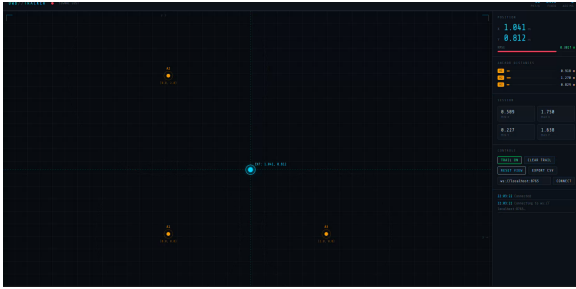
Analysis

The primary issue appears to originate from inaccurate range measurements rather than the trilateration algorithm itself. Since trilateration directly depends on the measured ranges, biased ranges result in distorted position estimates. The observed instability in the Y direction suggests that Anchor 2 may be contributing disproportionately to the localization error. EKF instability was also observed; however, it is likely a secondary effect caused by biased input measurements.

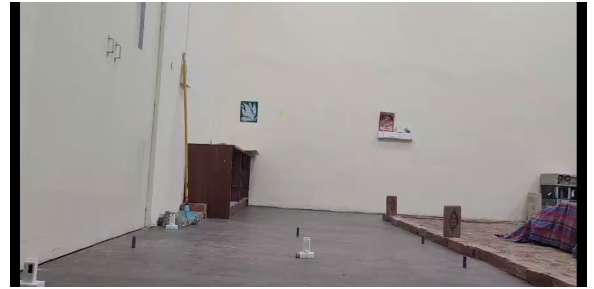
Conclusions

- Range measurement bias is the dominant source of localization error.
- Trilateration is functioning as expected given the provided range values.
- Anchor 2 should be investigated for calibration or hardware-related issues.
- Improving antenna delay calibration should be the highest priority.
- Addition of a fourth anchor is expected to improve overall localization accuracy and robustness.

Reference Images

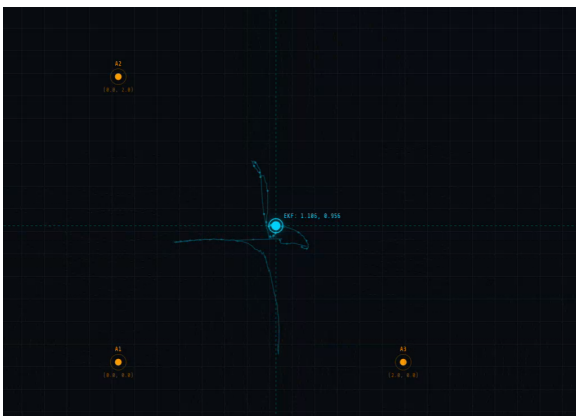


(a) GUI: Tag Positioned at Centre

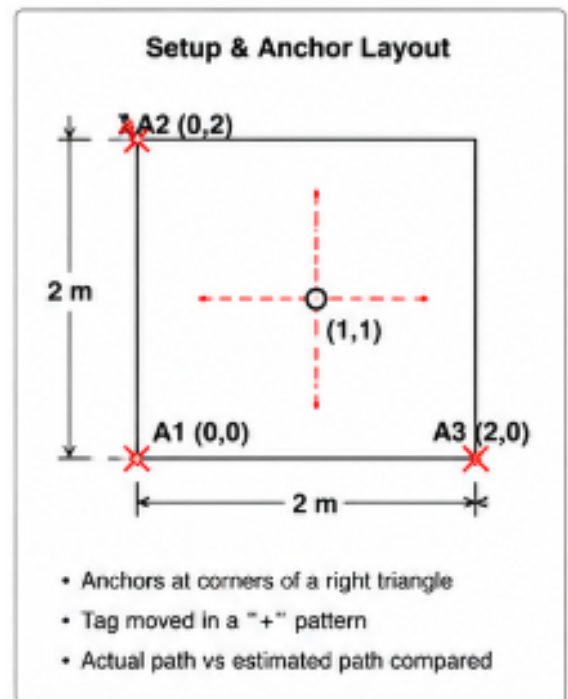


(b) Setup: Tag Positioned at Centre

Figure 1: UWB localization GUI and experimental setup with the tag positioned at the centre.



(a) GUI: Obtained Pattern



(b) Attempted Pattern

Figure 2: UWB localization GUI and attained and attempted pattern.